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#include <stdio.h>
#define MAX 5

int queue[MAX];
int front = -1, rear = -1;

void enqueue()
{
    int value;

    if (rear == MAX - 1)
    {
        printf("Queue Overflow\n");
        return;
    }

    printf("Enter the element: ");
    scanf("%d", &value);

    if (front == -1)
        front = 0;

    rear++;
    queue[rear] = value;

    printf("Element inserted successfully.\n");
}

void dequeue()
{
    if (front == -1)
    {
        printf("Queue Underflow\n");
        return;
    }

    printf("Deleted element = %d\n", queue[front]);

    if (front == rear)
    {
        front = rear = -1;
    }
    else
    {
        front++;
    }
}

void peek()
{
    if (front == -1)
    {
        printf("Queue is Empty\n");
    }
    else
    {
        printf("Front element = %d\n", queue[front]);
    }
}

void display()
{

```

```
int i;

if (front == -1)
{
    printf("Queue is Empty\n");
    return;
}

printf("Queue elements are: ");

for (i = front; i <= rear; i++)
{
    printf("%d ", queue[i]);
}

printf("\n");
}

int main()
{
    int choice;

    do
    {
        printf("\nQUEUE MENU\n");
        printf("1.Enqueue\n");
        printf("2.Dequeue\n");
        printf("3.Peek\n");
        printf("4.Display\n");
        printf("5.Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                enqueue();
                break;

            case 2:
                dequeue();
                break;

            case 3:
                peek();
                break;

            case 4:
                display();
                break;

            case 5:
                printf("Program Exited.\n");
                break;

            default:
                printf("Invalid Choice!\n");
        }

    } while (choice != 5);

    return 0;
}
```